RAG-Based Tools vs. General-Purpose LLMs for Academic Research

A Comparative Guide for Students

Executive Summary

Not all AI tools are created equal when it comes to academic research. RAG-based tools like Google NotebookLM restrict their answers to documents you provide and cite every claim back to a specific passage, making verification straightforward. General-purpose LLMs like ChatGPT, Claude, and Gemini draw on broad training data, which makes them flexible but also prone to fabricating sources and making unverifiable claims.

Neither category eliminates the researcher's responsibility to check every claim. RAG tools reduce hallucination risk and make citation checking easier, but they can still misread sources, overgeneralize, or cite passages that do not actually support the claim. General-purpose LLMs can be useful for brainstorming and explanation but should never be trusted for factual citations without independent verification.

How AI Language Models Actually Work

AI Is Probabilistic — Not Intelligent

Despite how impressive AI responses can appear, large language models are fundamentally **word-prediction machines**. When you type a prompt, the model does not understand your question the way a human would. Instead, it looks at the sequence of words you typed and calculates the probability of which word is most likely to come next, then the next, then the next — one word at a time.

Think of it like a very sophisticated autocomplete. When your phone suggests the next word in a text message, it is doing a simpler version of the same thing. An LLM does this at a massive scale — trained on billions of sentences, it has learned statistical patterns about which words tend to follow which other words in what contexts. It is matching patterns, not reasoning.

This means AI does not "know" facts. It has no memory, no beliefs, and no understanding. When it produces a correct answer, it is because the statistical patterns in its training data happen to lead to the right words in the right order. When it produces a wrong answer — or fabricates a source that does not exist — it is because the same word-prediction process generated plausible-sounding words that happen to be false. The model cannot tell the difference.

Non-Determinism — The Same Prompt Can Produce Different Results

Because AI is probabilistic, it introduces **randomness** into its word selection. At each step, instead of always picking the single most likely next word, the model randomly samples from among the top candidates. This randomness is controlled by a setting called **temperature** — higher temperature means more randomness, lower means more predictable output.

The practical consequence is **non-determinism**: if you type the exact same prompt twice, you may get two different answers. Both may be correct, both may be wrong, or one may be correct and the other fabricated. There is no guarantee of consistency.

This is fundamentally different from how a search engine or database works. When you search Google for "capital of France," you get the same result every time. When you ask an LLM, you will usually get "Paris," but the surrounding explanation, phrasing, and any additional "facts" may change — and some of those additional details may be fabricated on one run but not another.

Why this matters for research: You cannot assume that because an AI gave you a correct, well-cited answer once, it will give the same answer again. Every AI output must be independently verified against primary sources.

Key Definitions

What is a General-Purpose LLM?

A large language model trained on vast amounts of internet text. When you ask a question, it generates an answer by predicting one word after another based on statistical patterns learned during training. It does not look anything up in real time (unless it has a search feature enabled) and cannot reliably tell you where it got a specific fact. Because it is probabilistic, the same question may produce different answers on different occasions. Examples include ChatGPT, Claude, and the standard Gemini chat interface.

What is a RAG-Based Tool?

RAG stands for Retrieval-Augmented Generation. A RAG system works in stages: (1) it chunks your uploaded documents into passages, (2) it converts those passages into searchable vectors, (3) when you ask a question, it retrieves the most relevant passages, and (4) it generates an answer constrained to those passages, attaching a citation to each claim (Google Research). NotebookLM is the most prominent example.

What is Source Grounding?

Source grounding means the model is required to anchor its answers to user-provided documents rather than free-associating from training data. Claims in a NotebookLM answer typically carry a numbered citation that links to the relevant passage in your uploaded source. You can click any citation to jump to the original text and verify the claim yourself.

Comparison Table

Dimension	RAG-Based Tools (e.g., NotebookLM)	General-Purpose LLMs (e.g., ChatGPT, Claude)
How it works	Retrieves passages from your documents, then generates an answer constrained to those passages	Predicts the next word based on statistical patterns from training data — probabilistic, not deterministic
Knowledge source	Documents you upload and select	Broad training data from the internet
Grounding	Answers designed to be grounded in your selected sources	No source constraint; generates from learned patterns
Citations	Claims typically cite a specific passage you can click and verify	No reliable citations; may fabricate sources entirely
Consistency	More consistent because answers are anchored to specific passages	Non-deterministic: same prompt may produce different answers each time
Hallucination risk	Reduced but not eliminated (~13% unsupported vs ~40% for open chatbots)	Higher; fabricates facts, quotes, and references
"I don't know"	Will often say sources do not cover a topic	Tends to generate plausible-sounding answers even when uncertain
Best for	Summarizing readings, finding themes, checking comprehension, study aids	Brainstorming, explaining concepts, drafting outlines

Dimension	RAG-Based Tools (e.g., NotebookLM)	General-Purpose LLMs (e.g., ChatGPT, Claude)
Academic integrity	Lower risk if used for comprehension and organization	High risk if used to generate content presented as your own work

How Source-Grounded Systems Handle Hallucinations

RAG Reduces Hallucinations — But Does Not Eliminate Them

RAG systems mitigate hallucinations by retrieving real passages from your documents before answering and attaching a citation to each claim. A 2025 study found that RAG with reliable information sources significantly reduced hallucination rates — with GPT-4 showing 0% hallucination when using a curated source, compared to 6% when using open web results (PubMed). Another framework, MEGA-RAG, demonstrated over 40% reduction in hallucination rates compared to baseline models (PMC).

The Paradox: RAG Can Also Increase Hallucinations

A May 2025 study from Google Research found that while RAG generally improves performance, it paradoxically reduces the model's ability to abstain from answering when appropriate. When context is merely relevant but insufficient, models still try to answer. The model Gemma produced incorrect answers on 10.2% of questions with no context, but that jumped to 66.1% when given insufficient context (Google Research).

Garbage In, Garbage Out

If you add a low-quality, biased, or factually wrong source, NotebookLM will faithfully cite the bad information. It will not fact-check your sources. The quality of your source collection directly determines the quality of the output.

Citation Drift

The main failure mode occurs when a citation points to a passage that is related to the claim but does not actually support it. The citation number looks reassuring either way — only reading the passage tells you which case you have.

General-Purpose LLMs: Fabricated Citations

General-purpose LLMs have a well-documented tendency to fabricate citations — generating plausible-looking references to papers, authors, and journals that do not exist. American University warns that "any submitted work referring to fabricated sources may constitute a violation, including material resulting from generative AI hallucinations" (American University).

Data Privacy

Account Type	Human Review	Used for AI Training
Google Workspace for Education	Not reviewed by humans, even with feedback	Not used to train AI models
Google Workspace (work)	Same protections as Education	Not used to train AI models
Personal Google Account	Feedback may be reviewed; retained up to 3 years	Not used unless feedback is provided

Student Checklist: Safe, Caution, or High Risk

Use the tool when ALL of these are true:

✓ Your instructor has explicitly permitted AI use for this assignment

✓ You are using it for comprehension, organization, or study aids — not to generate submitted prose

✓ You click through and verify every citation on load-bearing claims

✓ You are not uploading confidential, unpublished, or restricted data

✓ You disclose your AI use if your institution requires it

✓ The final submitted work represents your own thinking, analysis, and writing

Use caution when ANY of these apply:

■ The tool is writing or rewriting prose you plan to submit

■ You are summarizing sources you have not actually read yourself

■ The tool is generating interpretations you cannot independently verify

■ You are working with unpublished data or confidential information

■ Your institution's AI policy or your instructor's rules are unclear

High risk — likely an academic integrity violation:

✗ AI completes graded work for you (essays, code, exams, problem sets)

✗ You include fabricated or unchecked citations generated by an AI tool

✗ AI replaces required reading or research you were supposed to do yourself

✗ You violate an explicit "no AI" rule in the syllabus

✗ You present AI-generated content as your own original work without disclosure

✗ You use AI during a closed-book exam without explicit permission

Key Takeaways

• **AI is probabilistic, not intelligent.** It predicts the next word based on statistical patterns. Correct and fabricated answers come from the same process.

• **AI is non-deterministic.** The same prompt can produce different answers each time. Every output must be independently verified.

• **RAG tools are not hallucination-free.** They reduce unsupported claims but roughly one in eight may still not be fully supported.

• **Citation exists does not mean citation supports.** Always click through and verify.

• **Privacy depends on account type.** Education accounts offer strong protections; personal accounts may involve human review.

• **You are always responsible for what you submit.** Fabricated citations and unauthorized AI use are violations at most institutions.

This guide is for educational purposes and reflects policies and research available as of August 2026. Always consult your institution's current policies and your instructor's guidance.

© Dr. Mark Taormino · College of Southern Nevada